Factors and outcomes associated with corneal edema and Haabs striae in primary congenital glaucoma

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PURPOSE To identify specific factors and outcomes associated with corneal edema and Haabs striae in primary congenital glaucoma (PCG).

METHODS The medical records of patients with PCG from 2011 to 2023 with >3 months' follow-up were reviewed retrospectively. Preoperative details and final outcomes were compared between eyes with and without corneal findings. The right eye of bilateral cases and the affected eye in unilateral cases were included.

RESULTS A total of 58 patients (104 eyes, 69% male) underwent initial angle surgery at an average age of 297 ± 368 (median, 134) days. Corneal edema and Haabs striae were present preoperatively in 72 (69%) eyes of 41 patients and 68 (65%) eyes of 39 patients, respectively. Patients with corneal edema presented at a younger age ($P < 0.0001$) and with shorter axial length ($P = 0.01$) than those without edema. Univariate analysis showed that corneal edema was associated with worse visual acuity at final follow-up (OR = 4.4; 95% CI, 1.2-25.3). Patients with Haabs striae were older than those without striae ($P = 0.04$). After angle surgery, corneal edema was present at 1 month in 71% (95% CI, 52-84), at 2 months in 26% (95% CI, 12-42), at 3 months in 16% (95% CI, 6-30), and at 1 year in 3% (95% CI, 0-13). Corneal opacification did not resolve in 4 eyes of 3 patients after >4 years of follow-up.

CONCLUSIONS In our study cohort, corneal edema resolved in the majority of PCG cases within 2-3 months of initial angle surgery but was associated with younger age at presentation and worse visual acuity at final follow-up. (J AAPOS 2024;28:103860)

Primary congenital glaucoma (PCG) is characterized by trabeculodysgenesis and presents between birth and 3 years of life with increased intraocular pressure (IOP).[1,2] The classic triad of symptoms of childhood glaucomas includes epiphora, photophobia, and blepharospasm. Increased IOP can cause impairment of corneal endothelial cell pump function and subsequent swelling of the corneal stroma and epithelium.[3] In addition, the scleral distensibility in infants and young children often leads to buphthalmos with axial lengthening; however, the stiffer Descemet membrane can rupture, resulting in linear or circular Haabs striae.[4] Corneal edema, therefore, is likely caused both by elevated IOP overcoming the endothelial pump and from the loss of Descemet membrane's barrier function.

Although both edema and Haabs striae are common in PCG, very few studies have evaluated associated factors and subsequent outcomes of these corneal findings in affected children.[1,2] In addition, although corneal edema usually clears after surgery and obtaining IOP control, there are limited data regarding the timing and trajectory of resolution.

Subjects and Methods

The medical records of patients with a diagnosis of PCG seen at the Ann & Robert H. Lurie Children's Hospital, Chicago, or the University of Michigan between 2011 and 2023 were reviewed retrospectively. PCG diagnosis was based on the definition of the World Glaucoma Association.[5] Included eyes had no evidence of anterior segment dysgenesis, such as congenital ectropion uvea, Axenfeld anomaly, Rieger anomaly, Peters anomaly, or aniridia. Subjects also had no systemic diseases associated with childhood glaucoma, such as Sturge Weber syndrome or neurofibromatosis. Additional inclusion criteria were >3 months of postoperative follow-up and availability of pre- and postoperative examination

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Table 1. Demographics of patients with primary congenital glaucoma (PCG)

Study parameter	PCG (n = 58)
Sex	
Male	40
Female	18
Race	
White	30
Black	16
Asian	1
Other or did not report[a]	11
Ethnicity	
Non-Hispanic	45
Hispanic	13
Age at presentation, days	297 ± 368
Median	134
Range	2-1496
Laterality	
Bilateral	46
Unilateral	12
Final follow-up, days	1832 ± 1140
Median	1550
Range	116-4152

[a]10 patients who either self-reported as "other" or did not report, did self-report as Hispanic ethnicity.

details, including corneal findings and operative reports. This study, deemed exempt from approval by the institutional review boards of both institutions, adhered to the tenets of the Declaration of Helsinki. Data collection was deidentified and compliant with the US Health Insurance Portability and Accountability Act of 1996.

Data collected included sex, race, ethnicity, age at presentation, and details of all glaucoma surgeries. IOP was measured by iCare (Revenue, Vantaa, Finland), Tono-Pen (Reichert, Depew, NY), or Goldmann applanation. Goldmann applanation was typically performed in children 5 years and older who were cooperative for testing. iCare tonometry was used in the clinic when Goldmann application was not possible because of age and/or cooperation. Tono-Pen was used during examinations under anesthesia in the operating room. For the preoperative examination, IOP was recorded from the presenting clinic visit, unless an IOP could not be obtained in the clinic due to poor cooperation. Axial length was determined by A scan at time of examination under anesthesia. At final examination, best-corrected visual acuity (BCVA), IOP, number of glaucoma medications, and cycloplegic refraction were collected. BCVA was assessed with age appropriate optotypes and then converted into logMAR. Corneal edema resolution was defined as no diffuse corneal edema or no edema that was not within or immediately adjacent to Haabs striae.

Analysis of one eye per patient included the right eye of patients with bilateral disease and the affected eye of patients with unilateral disease. Secondary analysis of all eyes was also performed. Statistical analysis using GraphPad Prism 10 (GraphPad Software Inc, La Jolla, CA) included Mann-Whitney test (comparison between groups), χ^2 test, and Kaplan-Meier survival curves with 95% confidence intervals. Univariate logistic regression analysis included calculation of β value, odds ratio, and likelihood ratio test (LRT). All tests were two-sided, and P values <0.05 were considered statistically significant.

Results

A total of 104 eyes of 58 patients with a diagnosis of PCG were included. Average age at presentation was 297 ± 368 days (median, 134; range, 2-1496) See Table 1. Patients with bilateral disease (n = 46) were significantly younger than those with unilateral PCG (n = 12, P = 0.02; eSupplement 1, available at jaapos.org).

Seventy-two (69%) eyes of 41 (71%) patients had corneal edema at presentation. Analysis of 1 eye per patient (n = 58, right eye of bilateral cases and affected eye of unilateral cases) demonstrated that patients with corneal edema (n = 41) were younger (140.1 ± 117 days vs 674 ± 483 days [P < 0.0001]) at presentation than those without edema (n = 17). Further, axial length (22.6 ± 2.1 mm vs 24.5 ± 2.6 mm [P = 0.01]) was shorter at

Table 2. Characteristics and outcomes of patients with primary congenital glaucoma who presented with or without corneal edema: only one eye per patient used in analysis

Study parameter	+Corneal edema [n = 41 eyes][a]	No corneal edema [n = 17 eyes][a]	P value
Age at presentation, days	140 ± 117; 113 (2-684)	674 ± 483; 513 (93-1496)	<0.0001
Initial IOP, mm Hg	31.8 ± 8.2; 33 (18-50)	27.1 ± 8.4; 24.5 (14-45)	0.07
Initial AL, mm	22.6 ± 2.1; 22.3 (16.8-28.5)	24.5 ± 2.6; 24.3 (20.0-29.5)	0.01
Haabs striae at presentation	27 (66)	12 (71)	1.00
No. surgeries for corneal edema to resolve	1.3 ± 0.5; 1.0 (1.0-3.0)	NA	
Total no. angle surgeries	1.4 ± 0.5; 1 (1-3)	1.2 ± 0.4; 1 (1-2)	0.54
Total degrees of angle treated	275.9 ± 90.3; 360 (180-360)	238.2 ± 80.4; 180 (150-360)	0.11
No. eyes requiring non-angle glaucoma surgery	12 (29)	5 (29)	1.00
Follow-up, days	1729 ± 1187; 1283 (116-4152)	2043 ± 1018; 1570 (721-3986)	0.26
Final logMAR BCVA	0.70 ± 0.68; 0.35 (0-3.0) [n = 30]	0.32 ± 0.38; 0.20 (0-1.18) [n = 15]	0.04
Final IOP, mm Hg	13.8 ± 4.1; 13 (5-25)	13.8 ± 3.6; 13 (9-20)	0.91
Final no. glaucoma medications	0.6 ± 0.9; 0 (0-3)	1.1 ± 1.4; 2 (0-4)	0.14
Final SE of cycloplegic refraction	−3.46 ± 4.22; −0.94 (−15.13 to 2.25)	−2.73 ± 4.9; −0.50 (−11.25 to 2.25)	0.25
Final cup:disk ratio	0.72 ± 0.20; 0.80 (0.25-0.99)	0.72 ± 0.20; 0.78 (0.30-0.99)	0.92

AL, axial length; *BCVA*, best-corrected visual acuity; *IOP*, intraocular pressure; *SE*, spherical equivalent.
[a]Results presented as mean ± standard deviation followed by median (range) or as number (%).

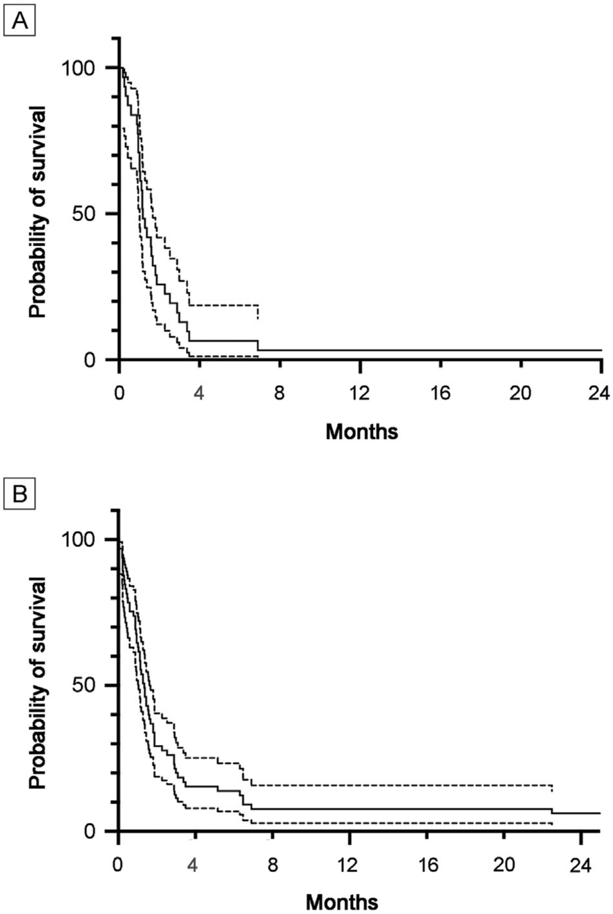

FIG 1. Rate of corneal edema resolution in primary congenital glaucoma. A, Kaplan Meier analysis of corneal edema resolution in 1 eye per patient (n = 58, right eye in bilateral cases or affected eye in unilateral cases), showed that 71% of eyes (95% CI, 52-84) had edema at 1 month, decreasing to 26% (95% CI, 12-42) at 2 months, 15% (95% CI, 5-29) at 3 months, 6% (95% CI, 1-18) at 6 months and 3% (95% CI, 0-13) at 1 year. B, Analysis of all eyes (n = 104) showed corneal edema rates of 64% (95% CI, 51-74) at 1 month, 29% (95% CI, 19-40) at 2 months, 22% (95% CI, 13-33) at 3 months, and 13% (95% CI, 6-21) at 6 months, 8% (95% CI, 3-16) at 1 year, and 6% (95% CI, 3-14) at 2 years.

presentation compared with eyes that did not have corneal edema. There was no difference ($P = 1.00$) in the percentage of eyes with Haabs striae among those with corneal edema (66%) versus those without corneal edema (71%). There was also no difference in the number of angle surgeries or total degrees of angle treated between eyes with and without edema (Table 2). Length of final follow-up did not differ between groups (1729 ± 1187 days vs 2043 ± 1018 days [$P = 0.26$]). Eyes that presented with corneal edema (n = 30) had worse final logMAR BCVA (0.70 ± 0.68) compared to eyes without edema (n = 15; 0.32 ± 0.38 [$P = 0.04$]), and univariate logistic regression analysis also showed an association between corneal edema and worse final vision (OR = 4.4; 95% CI, 1.2-25.3); see eSupplement Table 2 (available at jaapos.org). Thirteen patients were too young or uncooperative for optotype testing at final follow-up. At final follow-up, there was no difference in IOP, number of glaucoma medications, spherical equivalent of cycloplegic refraction, or cup:disk ratio. Analysis of all eyes (n = 104; see eSupplement 3), showed similar results.

Corneal edema resolved in 96% of eyes (100/104 eyes) in 95% of patients (55/58). Analysis of 1 eye per patient (n = 41) showed that the length of time from initial angle surgery to resolution of corneal edema was 49.2 ± 51.3 days (median, 31; range 1-210), compared with 61.2 ± 94.4 days in all eyes (median 38; range, 684). Unilateral cases resolved more quickly than bilateral cases ($P < 0.05$; see eSupplement 1). Additional glaucoma surgery (angle or non-angle) was performed if the IOP remained uncontrolled as indicated by a combination of elevated IOP measurements, continued corneal edema, increased axial length, myopic shift, or worsening optic nerve cupping. In the analysis of 1 eye per patient (right eye of bilateral cases or affected eye of unilateral cases), 31 had corneal edema resolution after one angle surgery and 9 showed resolution after two angle surgeries. Kaplan-Meier analysis of 1 eye per patient (Figure 1) showed that corneal edema was present in 71% of eyes at 1 month, 26% at 2 months, 16% at 3 months, and 3% at 1-year. Corneal edema resolved faster in eyes requiring 1 angle surgery compared to eyes needing 2 angle surgeries ($P = 0.03$). See eSupplements 4 and 5 (available at jaapos.org).

Corneal edema did not resolve in 4 eyes of 3 patients. One patient with nonresolving central edema of the left eye presented with PCG at birth. He underwent Descemet stripping automated endothelial keratoplasty at 6 months of age despite IOP control after undergoing 2 angle surgeries. At 4 years of age, the endothelial graft was clear, but BCVA in the affected eye was 20/200 (limited by amblyopia), compared with 20/40 in the other eye. In a second patient, the corneal edema in the left eye did not resolve despite IOP control after a single angle surgery. However, the edema was outside the visual axis, and at 5 years of age the BCVA was 20/70 (compared with 20/25 in the other eye). Both eyes of 1 patient had severe corneal edema, with diffuse stromal scarring at birth, but corneal surgery was not pursued after obtaining IOP control because of extensive optic neuropathy in both eyes and social constraints. At 9 years of age, his BCVA was counting fingers in both eyes.

Sixty-eight (65%) eyes of 39 (67%) patients had Haabs striae at presentation. Analysis of 1 eye per patient (Table 3) showed that in contrast to corneal edema, patients with eyes with Haabs striae were older (347 ± 405 days) at presentation than those without Haabs striae (193 ± 254 days [$P = 0.04$]). There was no difference in IOP at presentation, number of angle surgeries, or total degrees of angle treated. In addition, there was no significant difference ($P = 1.00$) in percentage of eyes with or without Haabs striae that had corneal edema at presentation. In the

Table 3. Characteristics and outcomes of patients with primary congenital glaucoma who presented with or without Haabs striae: only one eye per patient used in analysis

Study parameter	+Haabs striae [39 eyes][a]	No Haabs striae [19 eyes][a]	*P* value
Age at presentation, days	347 ± 405; 148 (2-1496)	193 ± 254; 111 (18-1125)	0.04
Initial IOP, mm Hg	31.4 ± 8.7; 31 (14-50)	28.4 ± 7.8; 26.5 (18-40)	0.23
Initial AL, mm	23.6 ± 2.5; 23.2 (19.0-29.5)	22.3 ± 2.2; 22.5 (16.8-26.6)	0.11
Corneal edema at presentation	27 (67)	14 (74)	1.00
Total no. angle surgeries	1.4 ± 0.5; 1 (1-3)	1.2 ± 0.4; 1 (1-2)	0.37
Total degrees of angle treated	272.3 ± 88.2; 300 180-360	249.5 ± 89.5; 180 (150-360)	0.36
No. eyes requiring non-angle glaucoma surgery	12 eyes (31)	5 eyes (26)	1.00
Follow-up, days	1889 ± 1126; 1550 (116-4152)	1678 ± 1188; 1555 (120-3986)	0.61
Final logMAR BCVA	0.62 ± 0.67; 0.45 (0.0-3.0) [n = 32]	0.41 ± 0.42; 0.25 (0-1.3) [n = 13]	0.36
Final IOP, mm Hg	13.8 ± 3.8; 13 (7-25)	13.8 ± 4.3; 13.5 (5-21)	0.96
Final no. glaucoma medications	0.7 ± 1.1; 0 (0-4)	0.8 ± 1.0; 0 (0-3)	0.60
Final SE of cycloplegic refraction	−3.33 ± 4.32; −2.81 (−15.13 to 2.25)	−3.01 ± 4.76; −0.88 (−14.25 to 1.50)	0.70
Final cup:disk ratio	0.74 ± 0.19; 0.8 (0.25-0.99)	0.68 ± 0.21; 0.70 (0.30-0.99)	0.31

AL, axial length; *BCVA*, best-corrected visual acuity; *IOP*, intraocular pressure; *SE*, spherical equivalent.
[a]Results presented as mean ± standard deviation followed by median (range) or as number (%).

secondary analysis of all eyes, axial length at presentation was longer in eyes with Haabs striae, but the difference was not significant in the analysis of 1 eye per patient. See eSupplement 6 (available at jaapos.org). At final follow-up, there was no difference in BCVA, IOP, number of glaucoma medications, spherical equivalent, or cup to disc ratio between eyes with and without Haabs striae.

Discussion

Corneal edema and Haabs striae are common presenting symptoms in PCG and other forms of childhood glaucomas. Even after achieving IOP control, Haabs striae and the possible resulting corneal scarring from edema can continue to affect visual function.[4,6,7] However, few studies have focused on the prevalence of these findings and identified associated factors and outcomes.

Corneal edema in the context of high IOP is due to impairment of endothelial cell pump function, resulting in excess fluid trapped within the stroma and epithelium.[3] In the current study, the prevalence of corneal edema in PCG was almost 70%, which was similar to other reports.[8,9] We found that the average and median age of presentation was between 3 and 5 months in eyes with edema but >1 year for eyes without edema. This may be due to differences in disease pathogenesis. More severe trabeculodysgenesis or absence of scleral collecting channels can cause high IOP pre- and perinatally, resulting in cloudy corneas at birth or soon thereafter.[10,11] Milder structural abnormalities within the aqueous outflow system could have fewer effects on endothelial cell function and cause more mild disease with later onset.[12] This association between corneal edema and younger age is important, because it may lead to delayed diagnosis of PCG in children >1 year of age.

Although most specialists who take care of children with glaucoma anecdotally know that corneal edema resolves in the majority of cases, the rate of clearance and the actual percentage is underreported. In the current study, IOP control was achieved in all patients, although 17 required non-angle surgery (most commonly glaucoma drainage device placement) in addition to the primary angle surgeries. The average and median time from initial angle surgery until documented resolution of corneal edema was between 1 and 2 months, with approximately 90% of eyes having no edema by 6 months. This was a higher rate of resolution compared to an Indian study in which almost 40% of eyes still had edema at final follow-up (median, 17 months).[8] This difference may be due to specific genotypes that are prevalent in the Indian population that are associated with worse PCG phenotypes. These patients present at a younger age and have a worse response to angle surgery.[11]

To our knowledge, no other reports from other populations directly assess corneal edema resolution. However, change in central corneal thickness (CCT), which may reflect improvement in corneal edema, has been studied as a marker for monitoring IOP control postoperatively in children with glaucoma.[13,14] Similar to the rate of corneal edema resolution, CCT decreased within 2 weeks and on average stabilized by 3 months after surgery.[13] Our retrospective study did not include CCT data, because this was not routinely collected either pre- or postoperatively, but determination of corneal clarity relied on slit lamp examination, clarity of the red reflex on retinoscopy, and the ability to clearly visualize the posterior segment on funduscopy. Nevertheless, these results support observation of the corneal edema for at least 3-6 months after initial angle surgery.

Although corneal edema typically resolves, visually significant opacification may remain in a small percentage of PCG eyes due to decreased endothelial cell density or stromal scarring.[15] In some cases, endothelial keratoplasty can be an effective treatment, although more challenging in infants and young children compared to adults.[16-20] In eyes with stromal scarring, penetrating keratoplasty (PKP) is likely the better option despite having a higher graft failure rate in children compared to adults.[20-22] As a result, corneal surgeries may not be attempted until adolescence or adulthood, but the visual potential of the eye at that point may be severely affected by amblyopia.[16,17]

The presence of corneal edema at initial examination was associated with worse visual acuity at final follow-up in our study. This was despite no difference in the number of angle surgeries, the degrees of angle treated, the percent of eyes requiring non-angle glaucoma surgery (eg, glaucoma drainage device placement or trabeculectomy), or the final IOP. A few other studies have also shown that in PCG corneal edema at presentation was associated with worse visual acuity, however, these studies had a greater percentage of patients in whom corneal edema did not resolve.[8,18] Even with resolution, the worse outcomes could be due to other factors, such as dense amblyopia or more severe optic neuropathy, although we found no difference in cup:disk ratio at final follow-up. Additional studies are needed to better correlate corneal edema and optic nerve function (especially through methods such as optical coherence tomography) with axial length in pediatric glaucomas.

Unlike corneal edema, Haabs striae do not resolve with IOP control and can affect vision due to mild stromal haze obscuring the pupillary axis or by their causing significant irregular astigmatism.[4] We found that patients presenting with Haabs striae were significantly older than those without. Although at time of presentation, Haabs striae may be difficult to detect because of corneal edema, there was no significant difference in percentage of patients with corneal edema between those with or without Haabs striae. We found no difference in final visual acuity between patients with and without Haabs striae. Similarly, Patil and colleagues[4] also found that Haabs striae were not associated with worse BCVA in children with PCG as well as other secondary forms of glaucoma. Previous studies have shown no correlation between corneal edema and the presence of Haabs striae, which represent globe enlargement and breaks in Descemet's membrane.[4,19] While other studies comment on Haabs striae, there is limited published data in regards to their relation to buphthalmos or axial length.[4,7]

There are limitations to our study, including its retrospective nature and the number of patients. Given the young age of this patient population, optotype visual acuity was not able to be obtained in many patients, which may have influenced visual acuity outcomes. Furthermore, pachymetry and optical coherence tomography retinal nerve fiber layer thickness was not available for many patients, which could provide clinically useful information as well. In addition, there are different lengths of follow-up, and the findings are contingent upon accurate input of data, especially in regard to corneal edema resolution in young children. The time to corneal edema resolution is likely overestimated, as it required a documented exam with this finding. A strength of this study is that it focused on PCG and did not include secondary forms of glaucoma, which may have inherent corneal abnormalities that compound the edema or Haabs striae.

In summary, we found that corneal edema and Haabs striae were found in the majority of PCG patients in our study cohort at presentation. Patients with corneal edema presented at a younger age and shorter axial lengths compared with patients without corneal edema, and corneal edema correlated with worse BCVA at final follow-up. Corneal edema resolved in more than 95% of eyes with PCG, but this resolution may take up to 6 months. Understanding the effect of corneal edema and Haabs striae on overall visual prognosis is important in guiding clinical management in children with PCG.

Literature Search

A comprehensive literature search in PubMed was conducted on July 27, 2023, using the following terms: *primary congenital glaucoma*, *childhood glaucoma*, *corneal edema*, *Haabs striae*, *Descemet tears*, *pediatric penetrating keratoplasty*, and *pediatric endothelial keratoplasty*.

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